

A CASE-STUDY BASED REPORT

ON

**SPAM EMAIL CLASSIFICATION USING DECISION TREES &
TF-IDF**

Submitted in partial fulfilment of the requirements for the course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

Academic Year: 2026–27

Submitted by

P. Revanth — 2520030292

P. Sasi — 2520030505

Faculty Guide: [Add faculty guide name]

Department / Institution: [Confirm department and institution]

Contents

1. Abstract
2. 1. Introduction
3. 2. Problem Statement and Objectives
4. 3. Literature Review
5. 4. Dataset and Exploratory Data Analysis
6. 5. Methodology and Implementation
7. 6. Results and Evaluation
8. 7. Discussion
9. 8. Limitations and Future Work
10. 9. Conclusion
11. References

Abstract

Spam messages are unsolicited communications that may advertise products, promote fraudulent offers, or attempt to trick recipients into sharing information. This case study explores a machine-learning workflow for classifying text messages as spam or ham (legitimate). The workflow uses Term Frequency–Inverse Document Frequency (TF-IDF) to convert text into numerical features and a Decision Tree classifier to predict the message category.

The experiments use the UCI SMS Spam Collection dataset, containing 5,572 labelled messages: 4,825 ham and 747 spam. The project notebook reports stratified five-fold cross-validation results of 0.9607 ± 0.0059 accuracy, 0.9102 ± 0.0128 precision, 0.9242 ± 0.0184 recall, and 0.9167 ± 0.0127 F1-score. A separately reported confusion matrix is $\begin{bmatrix} 934 & 32 \\ 21 & 128 \end{bmatrix}$ for the test set. Since the available dataset contains SMS messages rather than email messages, this is an important scope limitation: the results demonstrate SMS text classification and do not by themselves establish performance on real email inboxes.

1. Introduction

Unwanted messages are a common problem across email and mobile messaging platforms. Manually checking every message is time-consuming, so text classification can help identify likely spam and prioritize messages for review. Machine-learning classifiers learn patterns from labelled examples, while text vectorization methods convert words and phrases into numeric representations that an algorithm can process.

This project focuses on a supervised-learning pipeline that combines TF-IDF features with a Decision Tree classifier. TF-IDF assigns higher importance to terms that are informative within a document while reducing the influence of terms that occur frequently across many documents. A Decision Tree makes predictions through a sequence of feature-based decision rules.

The project title uses the term “Email Classification,” but the dataset used in the implemented notebook is the UCI SMS Spam Collection. Therefore, the report describes the actual experiment as SMS spam classification and treats email classification as the intended application area rather than a directly validated result.

2. Problem Statement and Objectives

Problem statement: Given a labelled collection of short text messages, build and evaluate a machine-learning model that predicts whether a new message is spam or ham using TF-IDF text features and a Decision Tree classifier.

Objectives: (1) inspect the dataset and its class distribution; (2) represent message text numerically with TF-IDF; (3) train a Decision Tree classifier; (4) evaluate performance with cross-validation and a confusion matrix; (5) demonstrate predictions on example messages; and (6) document limitations and future improvements.

3. Literature Review

Paper 1 — Sjarif et al. (2019), “SMS Spam Message Detection using Term Frequency-Inverse Document Frequency and Random Forest Algorithm.” This open-access paper studies TF-IDF feature extraction with a Random Forest classifier for SMS spam detection. The authors report 97.50% accuracy in their experiments. It is relevant because it uses TF-IDF on SMS messages, although its classifier is Random Forest rather than the Decision Tree used in this project. Its reported score should not be compared directly with this project’s score unless the same data split, preprocessing, and evaluation procedure are used.

Paper 2 — Almeida, Gómez Hidalgo, and Yamakami (2011), “Contributions to the Study of SMS Spam Filtering: New Collection and Results.” This ACM conference paper introduces a public, non-encoded SMS spam collection and compares several machine-learning classifiers. The authors report that Support Vector Machines performed better than the other evaluated classifiers in their experiments. This work is particularly relevant because the SMS Spam Collection became a widely used benchmark for research on short-message spam filtering.

Together, these studies support the use of labelled SMS datasets and machine-learning text classification for spam detection. They also show that model performance depends on the classifier and experimental setup. The present project differs by using a Decision Tree classifier with TF-IDF features and by reporting cross-validation metrics and a confusion matrix.

4. Dataset and Exploratory Data Analysis

The dataset used by the notebook is the UCI SMS Spam Collection, loaded into a CSV file with two columns: label and message. The observed dataset shape is 5,572 rows and 2 columns. The class distribution is 4,825 ham messages and 747 spam messages. This means the data is imbalanced: ham messages make up approximately 86.6% of the collection and spam messages approximately 13.4%.

Exploratory data analysis (EDA) checked the dataset structure and class counts and generated a class-distribution chart. The chart makes the imbalance visible. Because the majority class is ham, accuracy alone is not sufficient to understand spam detection quality; precision, recall, F1-score, and the confusion matrix are also useful.

5. Methodology and Implementation

The implemented workflow can be summarized as follows: load the labelled CSV; separate message text and labels; convert message text into TF-IDF feature vectors; train a Decision Tree classifier; evaluate using stratified five-fold cross-validation; inspect the confusion matrix from the reported test evaluation; and run example messages through the trained model.

TF-IDF: Each message is transformed into a sparse numeric vector. The values reflect a term's importance in a message relative to its frequency across the collection. This allows text to be used as input to a conventional machine-learning model.

Decision Tree: The classifier learns decision rules from the training examples. Decision Trees are relatively easy to explain, but they can overfit if the tree is allowed to become too complex. Performance can depend on settings such as tree depth and minimum samples per leaf.

The notebook saved the trained model artifact to `/content/spam_decision_tree_model.pkl` in the Colab environment. That path is specific to the runtime; it is not automatically a file in the GitHub repository.

6. Results and Evaluation

The notebook reports stratified five-fold cross-validation results as follows. The $\pm$ values are the reported variation across folds.

Accuracy: 0.9607 ± 0.0059

Precision: 0.9102 ± 0.0128

Recall: 0.9242 ± 0.0184

F1-score: 0.9167 ± 0.0127

The separately shown confusion matrix is $\begin{bmatrix} 934 & 32 \\ 21 & 128 \end{bmatrix}$, with rows representing true ham/spam and columns representing predicted ham/spam. Therefore, 934 ham messages were correctly classified as ham, 32 ham messages were incorrectly predicted as spam, 21 spam messages were incorrectly predicted as ham, and 128 spam messages were correctly classified as spam.

The confusion matrix totals 1,115 examples. Its counts correspond to a test-set accuracy of approximately 95.25%. This is not the same as the cross-validation mean of 96.07%; the two results should be presented as separate evaluation outputs rather than merged into a single metric set.

Metric	Reported 5-fold CV result
Accuracy	0.9607 ± 0.0059
Precision	0.9102 ± 0.0128
Recall	0.9242 ± 0.0184
F1-score	0.9167 ± 0.0127

Confusion matrix (true labels by rows; predicted labels by columns):

	Predicted ham	Predicted spam
True ham	934	32

True spam	21	128
-----------	----	-----

7. Discussion

The cross-validation results suggest that the TF-IDF and Decision Tree pipeline can distinguish spam from ham reasonably well on the SMS dataset. The confusion matrix shows both types of error: false positives, where legitimate messages are flagged as spam, and false negatives, where spam messages are missed. In a real application, the relative cost of these errors should influence model selection and threshold or decision policies.

The dataset imbalance is important when interpreting results. A model could achieve high accuracy by favouring the majority class, so precision, recall, F1-score, and class-specific confusion-matrix counts should be reviewed together. Results from a published Random Forest or SVM study are useful context, but they are not a controlled head-to-head comparison with this Decision Tree model.

8. Limitations and Future Work

The most important limitation is the data-domain mismatch: the current dataset consists of SMS messages, not a corpus of full email messages with subject lines, sender information, headers, and longer body text. Therefore, the project should not claim that it has validated a real email spam filter.

Other limitations include possible sensitivity to the train/test split and Decision Tree settings, limited external validation, and the need to assess performance on newer or independently collected messages. Future work could test the pipeline on a public email dataset, compare Decision Tree against Naive Bayes, Logistic Regression, SVM, and Random Forest under the same split, tune tree complexity using cross-validation, and report per-class precision/recall and error examples.

9. Conclusion

This case study implemented a text-classification pipeline using TF-IDF and a Decision Tree to distinguish spam from legitimate messages in the UCI SMS Spam Collection. The notebook reports a five-fold cross-validation accuracy of 96.07% on average, alongside precision, recall, and F1-score, and provides a confusion matrix for a separate test evaluation. These results are promising for the tested SMS dataset, but they should not be described as proof of performance on actual email data. Testing on a suitable email dataset and comparing algorithms under consistent conditions are logical next steps.

References

- [1] N. N. A. Sjarif, N. F. Mohd Azmi, S. Chuprat, H. M. Sarkan, Y. Yahya, and S. M. Sam, "SMS Spam Message Detection using Term Frequency-Inverse Document Frequency and Random Forest Algorithm," *Procedia Computer Science*, vol. 161, pp. 509–515, 2019. DOI: 10.1016/j.procs.2019.11.150. Open paper: <https://www.sciencedirect.com/science/article/pii/S1877050919318617>
- [2] T. A. Almeida, J. M. Gómez Hidalgo, and A. Yamakami, "Contributions to the Study of SMS Spam Filtering: New Collection and Results," *Proceedings of the 11th ACM Symposium on Document Engineering*, pp. 259–262, 2011. DOI: 10.1145/2034691.2034742. Paper page/PDF: <https://dl.acm.org/doi/10.1145/2034691.2034742>
- [3] UCI Machine Learning Repository, "SMS Spam Collection." Dataset page: <https://archive.ics.uci.edu/dataset/228/sms+spam+collection>